

An Introduction to Schur–Weyl Duality

Thesis Draft

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Abstract

Schur–Weyl duality explains a remarkable compatibility between two apparently different kinds of symmetry. If V is a complex vector space and $V^{\otimes k}$ is its k -fold tensor power, then the general linear group $\mathrm{GL}(V)$ acts by changing every tensor factor in the same way, while the symmetric group S_k acts by permuting tensor positions. These two actions commute, and the resulting decomposition of $V^{\otimes k}$ is simultaneously controlled by irreducible representations of S_k and by polynomial representations of $\mathrm{GL}(V)$.

This thesis gives an introduction to the classical Schur–Weyl decomposition. The route is deliberately concrete: we begin with finite-dimensional representations, Young diagrams, Young symmetrizers, and Specht modules; then we construct Schur functors from tensor powers; finally we use the double-centralizer principle to prove the main theorem. Several low-degree examples are included to make the decomposition computable rather than merely formal. A final section sketches the connection with Schur polynomials and highest weights.

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1 Introduction

Tensor products are among the most useful constructions in linear algebra. They let one encode multilinear maps, build higher-degree functions from linear data, and turn a vector space V into a family of larger spaces

$$V, \quad V^{\otimes 2}, \quad V^{\otimes 3}, \quad \dots$$

When V is a complex vector space of dimension n , the tensor power $V^{\otimes k}$ has dimension n^k . At first sight this space looks too large to understand directly. Schur–Weyl duality says that it is large in a highly organized way.

There are two natural symmetries on $V^{\otimes k}$. First, every invertible linear map $g \in \mathrm{GL}(V)$ acts diagonally:

$$g \cdot (v_1 \otimes \cdots \otimes v_k) = (gv_1) \otimes \cdots \otimes (gv_k).$$

Second, the symmetric group S_k acts by permuting the tensor positions. In this draft we use a right action and the convention $(\sigma\tau)(i) = \sigma(\tau(i))$:

$$(v_1 \otimes \cdots \otimes v_k) \cdot \sigma = v_{\sigma(1)} \otimes \cdots \otimes v_{\sigma(k)}.$$

The central observation is that these two actions commute. Applying a linear change of coordinates to every tensor factor and then permuting the factors gives the same result as doing the permutation first and the linear change second.

The theorem proved in this thesis identifies the simultaneous decomposition that follows from this commuting pair of actions:

$$V^{\otimes k} \cong \bigoplus_{\lambda \vdash k, \ell(\lambda) \leq n} S^\lambda \otimes \mathbb{S}_\lambda(V).$$

Here $\lambda \vdash k$ means that λ is a partition of k , $\ell(\lambda)$ is the number of nonzero parts of λ , S^λ is the Specht module of the symmetric group S_k , and $\mathbb{S}_\lambda(V)$ is the Schur functor attached to λ . Thus the same partition λ labels an irreducible object on the symmetric-group side and an irreducible polynomial representation on the general-linear side.

The main goal is expository. We will not attempt to reproduce every proof from the representation theory of symmetric groups. Instead, we prove the basic facts needed to understand the theorem, state the standard classification theorem for Specht modules, prove the commuting action calculation in full, and then use a double-centralizer theorem as the proof spine. This follows the classical viewpoint of Schur and Weyl while keeping the presentation close to modern introductory references such as [FH91; Ful97; Eti+11].

Notation

Throughout this thesis:

- V denotes a finite-dimensional complex vector space;
- $n = \dim V$;
- k denotes the tensor-power degree;
- S_k denotes the symmetric group on $\{1, \dots, k\}$;
- $\lambda \vdash k$ means that λ is a partition of k ;
- S^λ denotes the Specht module indexed by λ ;
- $\mathbb{S}_\lambda(V)$ denotes the Schur functor indexed by λ .

All vector spaces and algebras are over $\mathbb{C}$ unless otherwise stated.

2 Background on Representations

This section recalls the representation-theoretic language used later. The main point is that finite-group representations over $\mathbb{C}$ split into irreducible pieces. Schur–Weyl duality is, in one sense, a very precise description of how such irreducible pieces occur inside a tensor power.

2.1 Representations and irreducibility

Definition 2.1. Let G be a group. A finite-dimensional representation of G is a finite-dimensional complex vector space W together with a group homomorphism

$$\rho : G \longrightarrow \mathrm{GL}(W).$$

We often write $g \cdot w$ instead of $\rho(g)w$.

Definition 2.2. A subspace $U \subseteq W$ is G -invariant if $g \cdot u \in U$ for every $g \in G$ and every $u \in U$. A nonzero representation W is irreducible if its only invariant subspaces are 0 and W .

Example 2.3. Let S_k act on $\mathbb{C}^k$ by permuting the standard basis vectors $e_1, \dots, e_k$. The line $\mathbb{C}(e_1 + \dots + e_k)$ is invariant. Its complement

$$\{(a_1, \dots, a_k) \in \mathbb{C}^k : a_1 + \dots + a_k = 0\}$$

is also invariant. For $k \geq 3$, this second representation is the standard representation of S_k .

Definition 2.4. If G acts on W , then the group algebra $\mathbb{C}[G]$ is the complex vector space with basis $\{e_g : g \in G\}$, with multiplication determined by $e_g e_h = e_{gh}$. A representation of G is equivalently a left $\mathbb{C}[G]$ -module.

The group algebra packages all group elements into one algebra. This is especially useful because the symmetric group action on $V^{\otimes k}$ extends linearly to an action of $\mathbb{C}[S_k]$.

2.2 Maschke's theorem and Schur's lemma

Theorem 2.5 (Maschke's theorem). *Let G be a finite group and let W be a finite-dimensional complex representation of G . Then W is a direct sum of irreducible representations.*

Proof. It is enough to show that every invariant subspace has an invariant complement. Let $U \subseteq W$ be invariant and choose any linear projection $p : W \rightarrow U$. This projection need not commute with the action of G , so we average it:

$$P(w) = \frac{1}{|G|} \sum_{g \in G} g \cdot p(g^{-1} \cdot w).$$

Then P is still a projection onto U , and the averaging makes it G -equivariant: $P(h \cdot w) = h \cdot P(w)$. Therefore $\ker P$ is G -invariant and

$$W = U \oplus \ker P.$$

Repeating this argument inductively on dimension decomposes W into irreducible summands. $\square$

Lemma 2.6 (Schur's lemma). *Let U and W be irreducible finite-dimensional representations of a group G . If $T : U \rightarrow W$ is G -equivariant, then either $T = 0$ or T is an isomorphism. In particular, if $U = W$, then every G -equivariant endomorphism of U is scalar.*

Proof. The kernel of T is an invariant subspace of U , and the image of T is an invariant subspace of W . Since U and W are irreducible, the only possibilities are $\ker T = U$, giving $T = 0$, or $\ker T = 0$ and $\operatorname{im} T = W$, giving an isomorphism.

For the last statement, let $T : U \rightarrow U$ be equivariant. Since $\mathbb{C}$ is algebraically closed, T has an eigenvalue α . Then $T - \alpha \operatorname{id}_U$ is equivariant and not injective, hence it is zero. $\square$

2.3 Characters and multiplicities

Definition 2.7. The character of a finite-dimensional representation $\rho : G \rightarrow \operatorname{GL}(W)$ is the function $\chi_W : G \rightarrow \mathbb{C}$ defined by

$$\chi_W(g) = \operatorname{tr}(\rho(g)).$$

Characters are important because they remember multiplicities. If

$$W \cong \bigoplus_i m_i U_i$$

is a decomposition into irreducible representations, then

$$\chi_W = \sum_i m_i \chi_{U_i}.$$

For finite groups over $\mathbb{C}$, irreducible characters form an orthonormal basis of the space of class functions. We will not need the full machinery of character theory, but the philosophy is helpful: representations can be understood by decomposing them into irreducibles and tracking how many times each one appears.

Definition 2.8. Let G act on W . The commutant, or centralizer algebra, of the action is

$$\text{End}_G(W) = \{T \in \text{End}(W) : T(g \cdot w) = g \cdot T(w) \text{ for all } g \in G, w \in W\}.$$

The commutant records the endomorphisms that are invisible to the group action. Schur–Weyl duality says that, for $W = V^{\otimes k}$, the endomorphisms invisible to the $\text{GL}(V)$ -action come from S_k , and the endomorphisms invisible to the S_k -action come from $\text{GL}(V)$.

3 Partitions, Young Diagrams, and Specht Modules

The irreducible representations of S_k are indexed by partitions of k . This section introduces the combinatorial objects that make that classification concrete.

3.1 Partitions and Young diagrams

Definition 3.1. A partition of k , written $\lambda \vdash k$, is a sequence

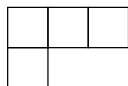
$$\lambda = (\lambda_1, \lambda_2, \dots, \lambda_r)$$

of positive integers such that

$$\lambda_1 \geq \lambda_2 \geq \dots \geq \lambda_r > 0, \quad \lambda_1 + \dots + \lambda_r = k.$$

The number r is the length of λ , denoted $\ell(\lambda)$.

We draw partitions using English Young diagrams: the first row has λ_1 boxes, the second row has λ_2 boxes, and so on. For example, the partition $(3, 1) \vdash 4$ has diagram



and the partition $(2, 1) \vdash 3$ has diagram



Definition 3.2. A Young tableau of shape $\lambda \vdash k$ is a filling of the boxes of the Young diagram of λ with the numbers $1, \dots, k$, each used once. The tableau is standard if its entries increase along each row from left to right and down each column from top to bottom.

Example 3.3. The tableau



is a standard tableau of shape $(2, 1)$. The tableau



is also standard. These two standard tableaux already hint at the fact that the Specht module $S^{(2,1)}$ has dimension 2.

3.2 Young symmetrizers

Fix a Young tableau T of shape $\lambda \vdash k$. Two subgroups of S_k are attached to T :

- the row group R_T , consisting of permutations that preserve each row of T as a set;
- the column group C_T , consisting of permutations that preserve each column of T as a set.

Definition 3.4. Inside the group algebra $\mathbb{C}[S_k]$, define

$$a_T = \sum_{\sigma \in R_T} \sigma, \quad b_T = \sum_{\tau \in C_T} \text{sgn}(\tau)\tau, \quad c_T = a_T b_T.$$

The element c_T is called a Young symmetrizer.

The element a_T symmetrizes along rows, while b_T antisymmetrizes along columns. The order matters. In many references one also considers $b_T a_T$, and the resulting modules are naturally isomorphic after the usual left-right convention is handled carefully.

Example 3.5. For the tableau

$$T = \begin{array}{|c|c|} \hline 1 & 2 \\ \hline 3 & \\ \hline \end{array},$$

the row group is $R_T = \{1, (12)\}$, while the column group is $C_T = \{1, (13)\}$. Hence

$$a_T = 1 + (12), \quad b_T = 1 - (13),$$

and

$$c_T = (1 + (12))(1 - (13)) = 1 - (13) + (12) - (132).$$

This element first alternates in the first column and then symmetrizes in the first row.

3.3 Specht modules

Definition 3.6. Let $\lambda \vdash k$, and choose a tableau T of shape λ . The Specht module S^λ is the irreducible S_k -module constructed from the Young symmetrizer c_T . Equivalently, it can be realized as the left module generated by a Young symmetrizer, after the standard convention for left and right actions is fixed.

The construction above depends on choosing a tableau, but the resulting irreducible representation depends only on the shape λ , up to isomorphism. This is one of the central results in the representation theory of the symmetric group.

Theorem 3.7 (Specht module classification). *The irreducible complex representations of S_k are indexed by partitions $\lambda \vdash k$. The representation corresponding to λ is the Specht module S^λ .*

We use this theorem as a standard input. A full proof is beautiful but long: it requires comparing Young symmetrizers, dominance order on partitions, and the structure of the group algebra $\mathbb{C}[S_k]$. For the purposes of Schur–Weyl duality, the important consequence is that the S_k -side of the decomposition of $V^{\otimes k}$ is completely indexed by partitions of k .

Theorem 3.8 (Hook-length formula). *For a box x in the Young diagram of λ , let $h(x)$ be its hook length: the number of boxes weakly to the right of x in the same row plus the number of boxes strictly below x in the same column. Then*

$$\dim S^\lambda = \frac{k!}{\prod_{x \in \lambda} h(x)}.$$

Example 3.9. For $\lambda = (2, 1)$, the hook lengths are

3	1
1	

so

$$\dim S^{(2,1)} = \frac{3!}{3 \cdot 1 \cdot 1} = 2.$$

The other partitions of 3 are (3) and $(1, 1, 1)$. Their Specht modules are the trivial and sign representations, both one-dimensional.

Exercise 3.10. Use the hook-length formula to compute the dimensions of the Specht modules for the five partitions of 4:

$$(4), \quad (3, 1), \quad (2, 2), \quad (2, 1, 1), \quad (1, 1, 1, 1).$$

4 Tensor Powers and Schur Functors

We now return to tensor powers. This section defines the two commuting actions on $V^{\otimes k}$, proves that they commute, and introduces Schur functors as the multiplicity spaces attached to Specht modules.

4.1 The two actions

Let V be a complex vector space of dimension n . The group $\mathrm{GL}(V)$ acts on $V^{\otimes k}$ by

$$g \cdot (v_1 \otimes \cdots \otimes v_k) = (gv_1) \otimes \cdots \otimes (gv_k).$$

The symmetric group S_k acts on the right by

$$(v_1 \otimes \cdots \otimes v_k) \cdot \sigma = v_{\sigma(1)} \otimes \cdots \otimes v_{\sigma(k)}.$$

This action extends linearly to an action of the group algebra $\mathbb{C}[S_k]$.

Proposition 4.1. *The left action of $\mathrm{GL}(V)$ and the right action of S_k on $V^{\otimes k}$ commute.*

Proof. It is enough to check pure tensors. Let $g \in \mathrm{GL}(V)$, let $\sigma \in S_k$, and let $v_1, \dots, v_k \in V$. Then

$$\begin{aligned} (g \cdot (v_1 \otimes \cdots \otimes v_k)) \cdot \sigma &= ((gv_1) \otimes \cdots \otimes (gv_k)) \cdot \sigma \\ &= (gv_{\sigma(1)}) \otimes \cdots \otimes (gv_{\sigma(k)}). \end{aligned}$$

On the other hand,

$$\begin{aligned} g \cdot ((v_1 \otimes \cdots \otimes v_k) \cdot \sigma) &= g \cdot (v_{\sigma(1)} \otimes \cdots \otimes v_{\sigma(k)}) \\ &= (gv_{\sigma(1)}) \otimes \cdots \otimes (gv_{\sigma(k)}). \end{aligned}$$

The two expressions agree, so the actions commute. □

The situation can be summarized by the following commuting square:

$$\begin{array}{ccc}
V^{\otimes k} & \xrightarrow{\text{right action of } S_k} & V^{\otimes k} \\
\downarrow \text{left action of } \text{GL}(V) & & \downarrow \text{left action of } \text{GL}(V) \\
V^{\otimes k} & \xrightarrow{\text{right action of } S_k} & V^{\otimes k}.
\end{array}$$

4.2 Definition of the Schur functor

Because $V^{\otimes k}$ is a right $\mathbb{C}[S_k]$ -module and S^λ is a left $\mathbb{C}[S_k]$ -module, we may form their tensor product over $\mathbb{C}[S_k]$.

Definition 4.2. For $\lambda \vdash k$, define the Schur functor

$$\mathbb{S}_\lambda(V) := V^{\otimes k} \otimes_{\mathbb{C}[S_k]} S^\lambda.$$

There are several equivalent descriptions. One can also view $\mathbb{S}_\lambda(V)$ as a multiplicity space:

$$\mathbb{S}_\lambda(V) \cong \text{Hom}_{S_k}(S^\lambda, V^{\otimes k}),$$

with the natural $\text{GL}(V)$ -action. Alternatively, after choosing a Young symmetrizer c_T , one can identify $\mathbb{S}_\lambda(V)$ with the image of c_T acting on $V^{\otimes k}$, up to harmless normalizing constants and left-right conventions.

Example 4.3. The partition (k) has one row. The corresponding Young symmetrizer symmetrizes all tensor positions, and

$$\mathbb{S}_{(k)}(V) \cong \text{Sym}^k(V).$$

The partition (1^k) has one column. The corresponding Young symmetrizer antisymmetrizes all tensor positions, and

$$\mathbb{S}_{(1^k)}(V) \cong \bigwedge^k V.$$

4.3 Basic properties

Proposition 4.4. *There are natural isomorphisms*

$$\mathbb{S}_{(k)}(V) \cong \text{Sym}^k(V), \quad \mathbb{S}_{(1^k)}(V) \cong \bigwedge^k V.$$

Proof. For (k) , the row group is all of S_k and the column group is trivial. The Young symmetrizer is therefore the full symmetrizer

$$\sum_{\sigma \in S_k} \sigma.$$

Its image consists exactly of symmetric tensors, which represent $\text{Sym}^k(V)$.

For (1^k) , the row group is trivial and the column group is all of S_k . The Young symmetrizer is the alternating operator

$$\sum_{\sigma \in S_k} \text{sgn}(\sigma) \sigma.$$

Its image consists exactly of alternating tensors, which represent $\bigwedge^k V$. □

Proposition 4.5. *If $\ell(\lambda) > n = \dim V$, then $\mathbb{S}_\lambda(V) = 0$.*

Proof. If $\ell(\lambda) > n$, then the Young diagram of λ has a column with more than n boxes. The Young symmetrizer defining $\mathbb{S}_\lambda(V)$ includes an antisymmetrization along this column. But there is no nonzero alternating tensor of degree $n + 1$ or higher in an n -dimensional vector space. Thus the corresponding image in $V^{\otimes k}$ is zero. $\square$

Remark 4.6. The length restriction $\ell(\lambda) \leq n$ is one of the main differences between the two sides of Schur–Weyl duality. Every partition of k gives a Specht module for S_k , but only partitions with at most n rows give nonzero Schur functors on an n -dimensional vector space.

5 Schur–Weyl Duality

The proof of Schur–Weyl duality is best understood through centralizers. The two actions on $V^{\otimes k}$ commute, so each action lands inside the centralizer of the other. Schur–Weyl duality says that these inclusions are as large as possible.

5.1 The double-centralizer principle

Theorem 5.1 (Double-centralizer theorem). *Let A be a finite-dimensional semisimple algebra over $\mathbb{C}$, and let W be a finite-dimensional A -module. Suppose*

$$W \cong \bigoplus_i U_i \otimes M_i$$

as an A -module, where the U_i are pairwise nonisomorphic irreducible A -modules and A acts on $U_i \otimes M_i$ only through U_i . Then

$$\mathrm{End}_A(W) \cong \bigoplus_i \mathrm{End}(M_i).$$

Moreover, the centralizer of $\mathrm{End}_A(W)$ inside $\mathrm{End}(W)$ is

$$\bigoplus_i \mathrm{End}(U_i),$$

acting on the U_i -factors.

Proof. By Schur’s lemma, an A -equivariant map from $U_i \otimes M_i$ to $U_j \otimes M_j$ must be zero when $i \neq j$. When $i = j$, the map must be the identity on U_i , up to scalar on the irreducible factor, tensored with an arbitrary linear map on M_i . This gives

$$\mathrm{End}_A(W) \cong \bigoplus_i \mathrm{End}(M_i).$$

Taking the centralizer once more exchanges the roles of the irreducible factors U_i and the multiplicity spaces M_i , giving the second statement. $\square$

This theorem is the algebraic reason that commuting actions produce a clean decomposition. One action decomposes the space into irreducibles; the other action is forced to operate on the multiplicity spaces.

5.2 The centralizer statement

Let

$$\rho : \mathrm{GL}(V) \rightarrow \mathrm{GL}(V^{\otimes k})$$

be the diagonal action, and let

$$\pi : \mathbb{C}[S_k] \rightarrow \mathrm{End}(V^{\otimes k})$$

be the linear extension of the place-permutation action. Since the two actions commute, we have inclusions

$$\rho(\mathrm{GL}(V)) \subseteq \mathrm{End}_{\mathbb{C}[S_k]}(V^{\otimes k}), \quad \pi(\mathbb{C}[S_k]) \subseteq \mathrm{End}_{\mathrm{GL}(V)}(V^{\otimes k}).$$

The classical centralizer theorem says that these inclusions become equalities if the left side is interpreted linearly:

$$\mathrm{span}_{\mathbb{C}}(\rho(\mathrm{GL}(V))) = \mathrm{End}_{\mathbb{C}[S_k]}(V^{\otimes k}), \quad \pi(\mathbb{C}[S_k]) = \mathrm{End}_{\mathrm{GL}(V)}(V^{\otimes k}).$$

We record this as a standard theorem. A full proof can be given using matrix coefficient calculations, polynomial density, or the representation theory of $\mathfrak{gl}(V)$.

Theorem 5.2 (Schur–Weyl centralizer theorem). *On $V^{\otimes k}$, the image of $\mathbb{C}[S_k]$ and the linear span of the image of $\mathrm{GL}(V)$ are mutual centralizers in $\mathrm{End}(V^{\otimes k})$.*

5.3 The main theorem

Theorem 5.3 (Schur–Weyl duality). *Let V be a complex vector space of dimension n . For every integer $k \geq 0$, there is an isomorphism of $S_k \times \mathrm{GL}(V)$ -modules*

$$V^{\otimes k} \cong \bigoplus_{\lambda \vdash k, \ell(\lambda) \leq n} S^\lambda \otimes \mathbb{S}_\lambda(V).$$

Furthermore, for each partition $\lambda \vdash k$ with $\ell(\lambda) \leq n$, the $\mathrm{GL}(V)$ -module $\mathbb{S}_\lambda(V)$ is irreducible.

Proof. By Maschke’s theorem, $V^{\otimes k}$, viewed as a representation of the finite group S_k , decomposes as a direct sum of irreducible S_k -modules:

$$V^{\otimes k} \cong \bigoplus_{\lambda \vdash k} S^\lambda \otimes M_\lambda,$$

where

$$M_\lambda \cong \mathrm{Hom}_{S_k}(S^\lambda, V^{\otimes k})$$

is the multiplicity space. The Specht classification tells us that the irreducible S_k -modules are exactly the S^λ .

Because the $\mathrm{GL}(V)$ -action commutes with the S_k -action, $\mathrm{GL}(V)$ acts on each multiplicity space M_λ . By definition, this multiplicity space is the Schur functor $\mathbb{S}_\lambda(V)$. Thus

$$V^{\otimes k} \cong \bigoplus_{\lambda \vdash k} S^\lambda \otimes \mathbb{S}_\lambda(V)$$

as a module for the commuting pair of actions.

If $\ell(\lambda) > n$, then $\mathbb{S}_\lambda(V) = 0$, as proved earlier. Hence only partitions with at most n parts remain. Finally, the double-centralizer theorem and the Schur–Weyl centralizer theorem imply that the nonzero multiplicity spaces for the S_k -decomposition are irreducible modules for the commuting $\mathrm{GL}(V)$ -action. Therefore $\mathbb{S}_\lambda(V)$ is irreducible whenever $\ell(\lambda) \leq n$. $\square$

Corollary 5.4. *As a representation of $\mathrm{GL}(V)$ alone, the multiplicity of $\mathbb{S}_\lambda(V)$ in $V^{\otimes k}$ is $\dim S^\lambda$.*

Proof. If we forget the S_k -action in

$$V^{\otimes k} \cong \bigoplus_{\lambda \vdash k, \ell(\lambda) \leq n} S^\lambda \otimes \mathbb{S}_\lambda(V),$$

then the factor S^λ becomes only a multiplicity space of dimension $\dim S^\lambda$. $\square$

6 Examples and Computations

The examples in this section illustrate how the theorem decomposes familiar tensor powers. They also show why the dimensions on both sides match.

6.1 Degree two

For $k = 2$, the partitions are (2) and $(1, 1)$. The corresponding Schur functors are

$$\mathbb{S}_{(2)}(V) = \mathrm{Sym}^2(V), \quad \mathbb{S}_{(1,1)}(V) = \bigwedge^2 V.$$

The two Specht modules are one-dimensional: $S^{(2)}$ is the trivial representation of S_2 , and $S^{(1,1)}$ is the sign representation. Therefore Schur–Weyl duality gives

$$V^{\otimes 2} \cong \mathrm{Sym}^2(V) \oplus \bigwedge^2 V$$

as a $\mathrm{GL}(V)$ -module.

This decomposition can be proved directly. Define

$$p_+(v \otimes w) = \frac{1}{2}(v \otimes w + w \otimes v), \quad p_-(v \otimes w) = \frac{1}{2}(v \otimes w - w \otimes v).$$

Then p_+ and p_- are projections,

$$p_+^2 = p_+, \quad p_-^2 = p_-,$$

and

$$p_+ + p_- = \mathrm{id}_{V^{\otimes 2}}.$$

The image of p_+ is $\mathrm{Sym}^2(V)$, and the image of p_- is $\bigwedge^2 V$. Hence

$$V^{\otimes 2} = \mathrm{im} p_+ \oplus \mathrm{im} p_-.$$

6.2 Degree three

For $k = 3$, the partitions are

$$(3), \quad (2, 1), \quad (1, 1, 1).$$

The Specht dimensions are

$$\dim S^{(3)} = 1, \quad \dim S^{(2,1)} = 2, \quad \dim S^{(1,1,1)} = 1.$$

If $n = \dim V \geq 3$, then all three Schur functors are nonzero, and Schur–Weyl duality gives, as a $\mathrm{GL}(V)$ -module,

$$V^{\otimes 3} \cong \mathrm{Sym}^3(V) \oplus \mathbb{S}_{(2,1)}(V)^{\oplus 2} \oplus \bigwedge^3 V.$$

If $n = 2$, the last term vanishes because $\bigwedge^3 V = 0$.

Example 6.1. Let $V = \mathbb{C}^3$. Then

$$\dim \mathrm{Sym}^3(\mathbb{C}^3) = \binom{3+3-1}{3} = \binom{5}{3} = 10,$$

and

$$\dim \bigwedge^3 \mathbb{C}^3 = 1.$$

Using the hook-content formula stated below,

$$\dim \mathbb{S}_{(2,1)}(\mathbb{C}^3) = 8.$$

Therefore

$$10 + 2 \cdot 8 + 1 = 27 = \dim(\mathbb{C}^3)^{\otimes 3}.$$

6.3 Dimension formulas

The Specht side has the hook-length formula. The general-linear side has a closely related formula.

Theorem 6.2 (Hook-content formula). *Let $\lambda \vdash k$ and $V = \mathbb{C}^n$. If $\ell(\lambda) \leq n$, then*

$$\dim \mathbb{S}_\lambda(\mathbb{C}^n) = \prod_{(i,j) \in \lambda} \frac{n + j - i}{h(i,j)},$$

where $h(i,j)$ is the hook length of the box in row i , column j .

Example 6.3. For $\lambda = (2, 1)$, the boxes are $(1, 1)$, $(1, 2)$, and $(2, 1)$. Their hook lengths are 3, 1, 1. Hence

$$\dim \mathbb{S}_{(2,1)}(\mathbb{C}^n) = \frac{n}{3}(n+1)(n-1) = \frac{n(n^2-1)}{3}.$$

For $n = 2$, this dimension is 2; for $n = 3$, it is 8.

Example 6.4. Take $V = \mathbb{C}^2$ and $k = 4$. Only partitions of 4 with at most two rows survive:

$$(4), \quad (3, 1), \quad (2, 2).$$

The hook-length formula gives

$$\dim S^{(4)} = 1, \quad \dim S^{(3,1)} = 3, \quad \dim S^{(2,2)} = 2.$$

The hook-content formula gives

$$\dim \mathbb{S}_{(4)}(\mathbb{C}^2) = 5, \quad \dim \mathbb{S}_{(3,1)}(\mathbb{C}^2) = 3, \quad \dim \mathbb{S}_{(2,2)}(\mathbb{C}^2) = 1.$$

Thus

$$1 \cdot 5 + 3 \cdot 3 + 2 \cdot 1 = 16 = \dim(\mathbb{C}^2)^{\otimes 4}.$$

6.4 A worked Young symmetrizer example

Let $\lambda = (2, 1)$ and choose

$$T = \begin{array}{|c|c|} \hline 1 & 2 \\ \hline 3 & \\ \hline \end{array}.$$

The Young symmetrizer is

$$c_T = 1 - (13) + (12) - (132).$$

Applying this to a pure tensor $u \otimes v \otimes w$ gives

$$(u \otimes v \otimes w)c_T = u \otimes v \otimes w - w \otimes v \otimes u + v \otimes u \otimes w - w \otimes u \otimes v.$$

The expression is symmetric in the first-row positions 1, 2 and alternating in the first-column positions 1, 3, according to the tableau. The span of such tensors is a concrete model for $\mathbb{S}_{(2,1)}(V)$, up to the conventional choice of left or right symmetrizer.

7 Characters and Highest Weights

The previous sections prove the structural decomposition. This final section briefly explains how the same representations are described by characters and highest weights. The point is not to develop the full theory of Lie algebras, but to show where the notation $\mathbb{S}_\lambda(V)$ comes from and why partitions are natural on the $\mathrm{GL}(V)$ -side as well.

7.1 Polynomial characters

Let $V = \mathbb{C}^n$, and let $g \in \mathrm{GL}_n(\mathbb{C})$ be diagonal with eigenvalues

$$x_1, \dots, x_n.$$

Then g acts diagonally on $V^{\otimes k}$, and its trace is

$$(x_1 + \dots + x_n)^k.$$

Schur–Weyl duality decomposes this character into irreducible polynomial characters:

$$(x_1 + \dots + x_n)^k = \sum_{\lambda \vdash k, \ell(\lambda) \leq n} (\dim S^\lambda) s_\lambda(x_1, \dots, x_n),$$

where s_λ is the Schur polynomial indexed by λ .

Definition 7.1. The Schur polynomial $s_\lambda(x_1, \dots, x_n)$ is the character of the $\mathrm{GL}_n(\mathbb{C})$ -representation $\mathbb{S}_\lambda(\mathbb{C}^n)$:

$$\chi_{\mathbb{S}_\lambda(\mathbb{C}^n)}(g) = s_\lambda(x_1, \dots, x_n)$$

when g has eigenvalues $x_1, \dots, x_n$.

This definition is representation-theoretic. There are also purely combinatorial definitions of Schur polynomials, for example as generating functions over semistandard Young tableaux. The equality between these viewpoints is one of the main bridges between representation theory and algebraic combinatorics.

7.2 Highest weights

Let $T \subseteq \mathrm{GL}_n(\mathbb{C})$ be the subgroup of diagonal matrices. A vector in a representation W has weight $\mu = (\mu_1, \dots, \mu_n) \in \mathbb{Z}^n$ if

$$\mathrm{diag}(t_1, \dots, t_n) \cdot w = t_1^{\mu_1} \cdots t_n^{\mu_n} w.$$

Polynomial irreducible representations of $\mathrm{GL}_n(\mathbb{C})$ are classified by dominant weights

$$\lambda_1 \geq \lambda_2 \geq \cdots \geq \lambda_n \geq 0.$$

These are exactly partitions with at most n parts.

Theorem 7.2. *The representation $\mathbb{S}_\lambda(\mathbb{C}^n)$ has highest weight $\lambda = (\lambda_1, \dots, \lambda_n)$, where trailing zeroes are added if necessary.*

This theorem places Schur–Weyl duality into the broader representation theory of reductive groups and Lie algebras. From that perspective, the condition $\ell(\lambda) \leq n$ is simply the condition that λ be a dominant polynomial weight for $\mathrm{GL}_n(\mathbb{C})$.

Example 7.3. For $V = \mathbb{C}^3$, the representation $\mathbb{S}_{(2,1)}(V)$ has highest weight $(2, 1, 0)$. Its dimension is 8, and it appears twice in $V^{\otimes 3}$ because the corresponding Specht module $S^{(2,1)}$ has dimension 2.

8 Conclusion

Schur–Weyl duality describes $V^{\otimes k}$ as a meeting point between two representation theories. The symmetric group S_k controls how tensor positions may be permuted, while the general linear group $\mathrm{GL}(V)$ controls how the vector space V itself may be changed. The actions commute, and the double-centralizer theorem explains why this produces a simultaneous decomposition.

The final decomposition

$$V^{\otimes k} \cong \bigoplus_{\lambda \vdash k, \ell(\lambda) \leq n} S^\lambda \otimes \mathbb{S}_\lambda(V)$$

summarizes the whole story. The same partition λ indexes a Specht module for S_k and a Schur functor for $\mathrm{GL}(V)$. The Specht dimension measures the multiplicity of the corresponding $\mathrm{GL}(V)$ -module, while the dimension of V determines which partitions survive.

Several natural directions remain beyond this introductory draft. One may study the Robinson–Schensted–Knuth correspondence, develop the full theory of symmetric functions, prove the complete highest-weight classification, or replace $GL(V)$ by orthogonal and symplectic groups, where Brauer algebras take the role played here by symmetric groups. These extensions all preserve the same guiding idea: tensor powers become understandable when their commuting symmetries are studied together.

References

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